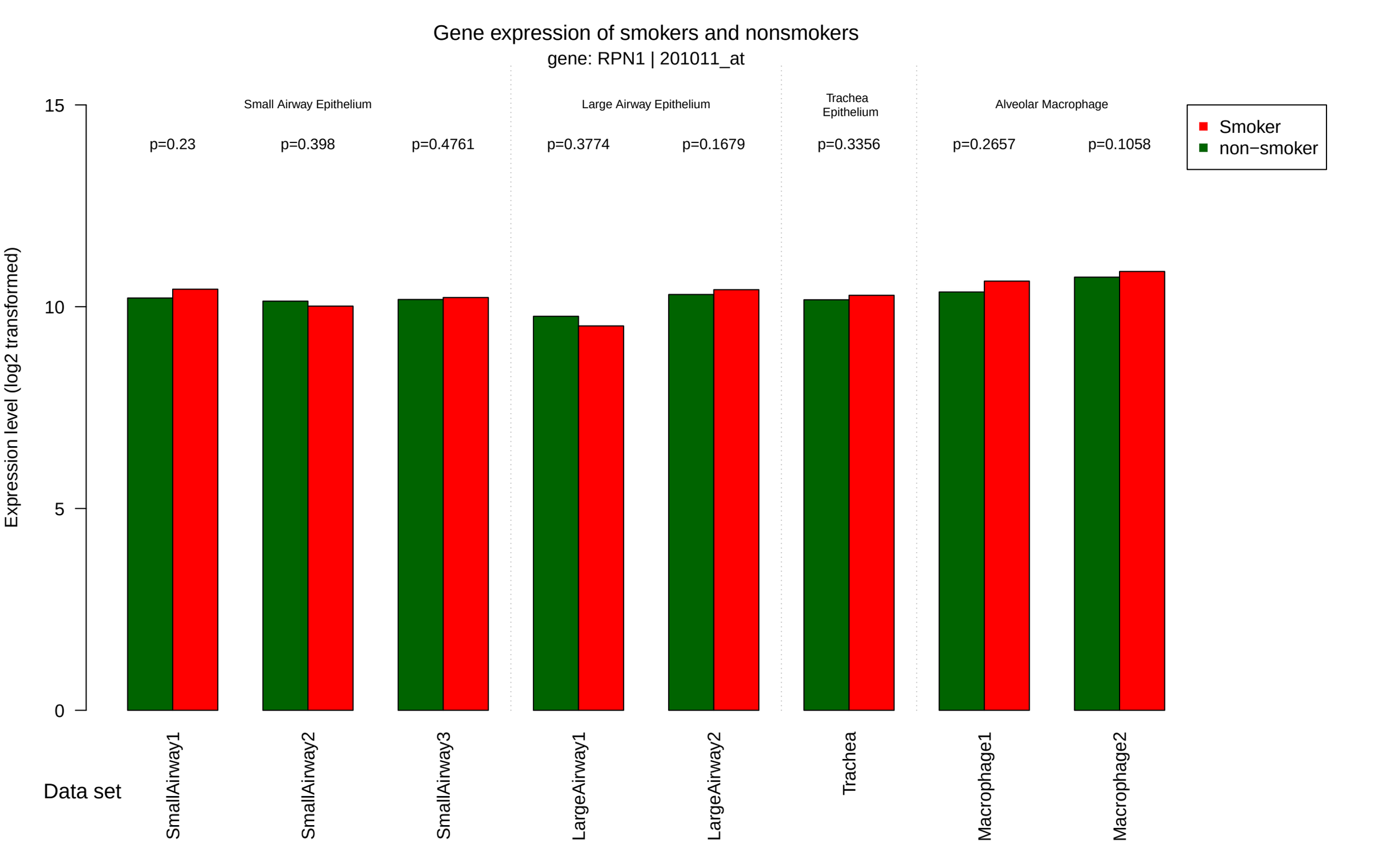
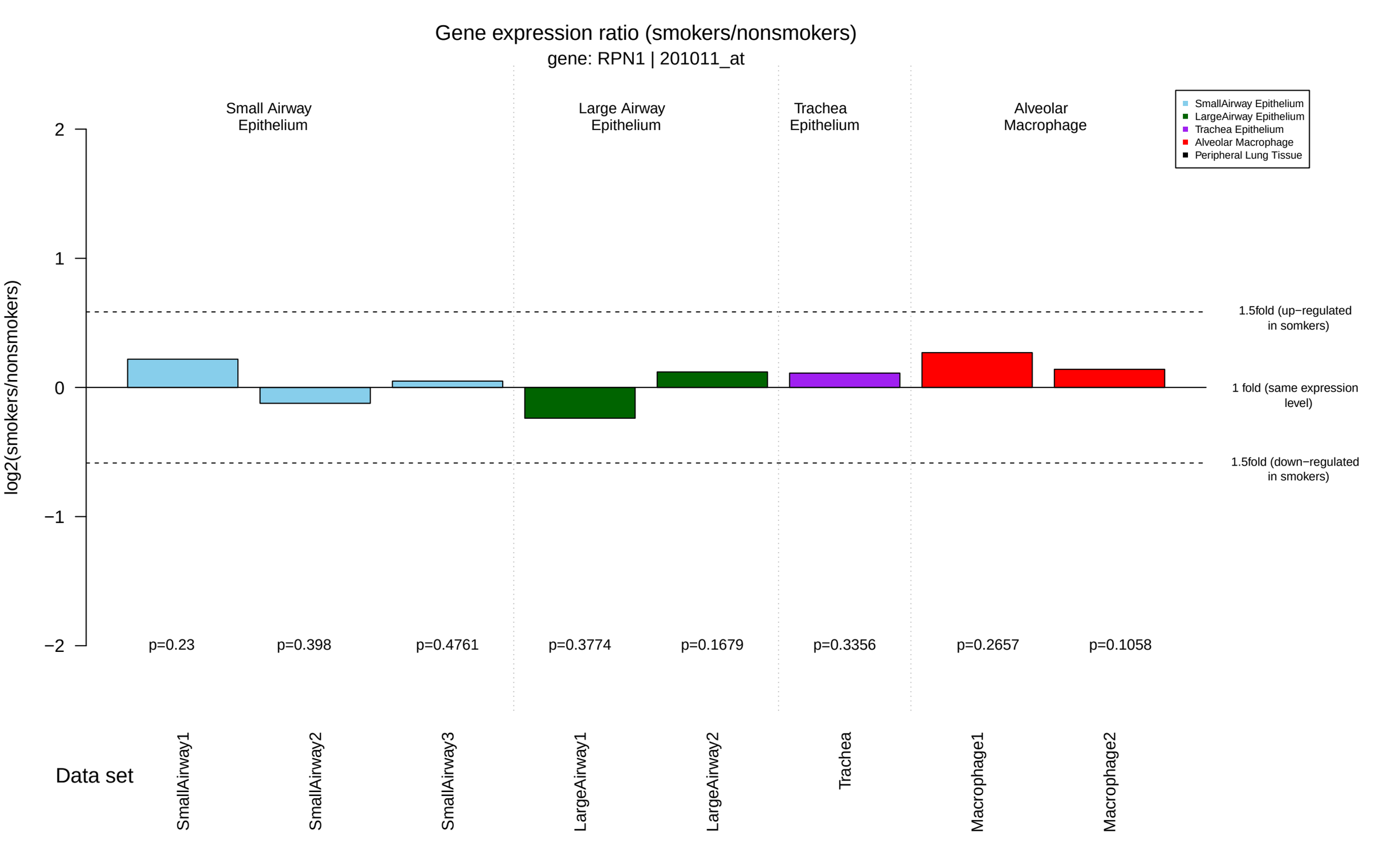
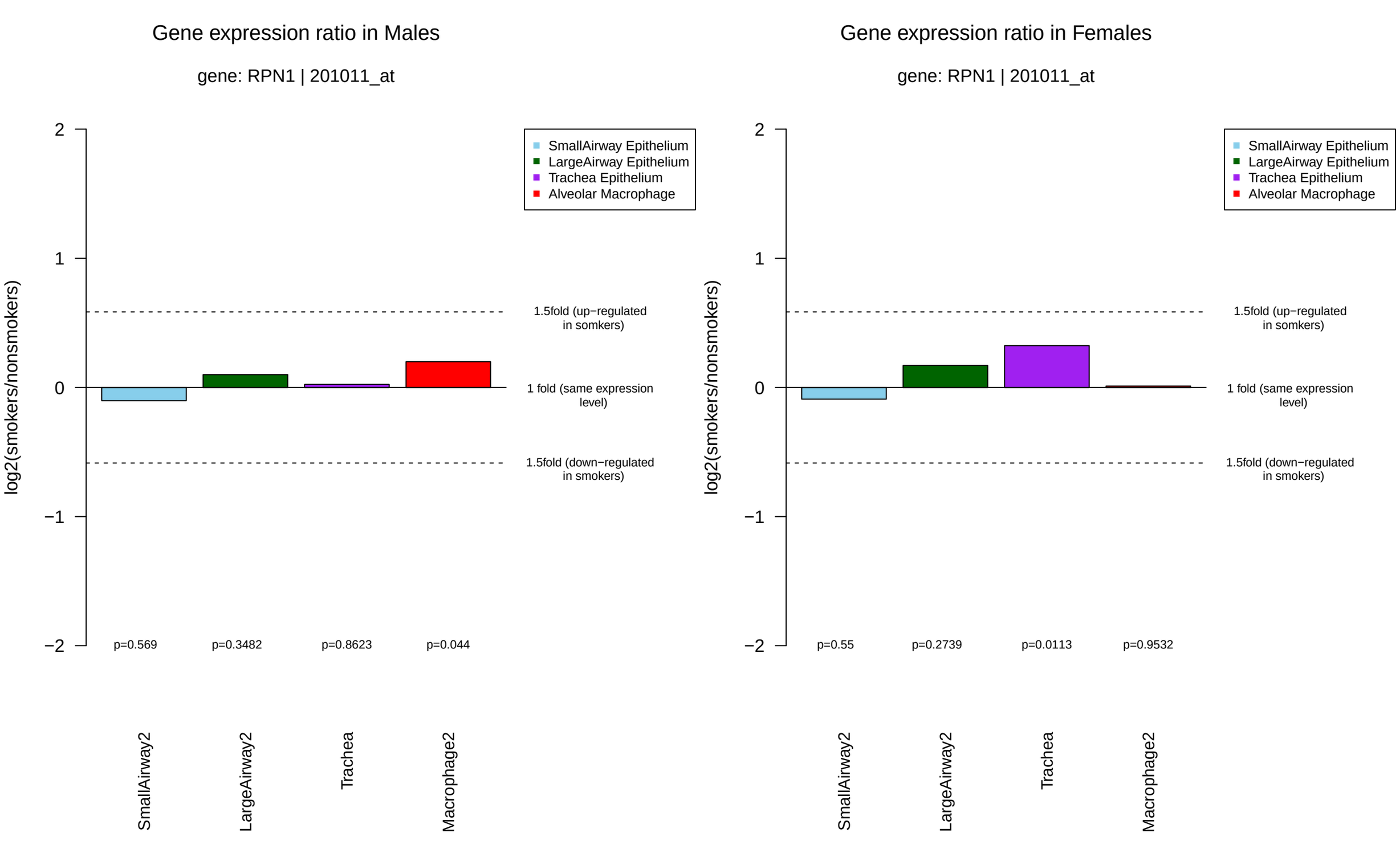
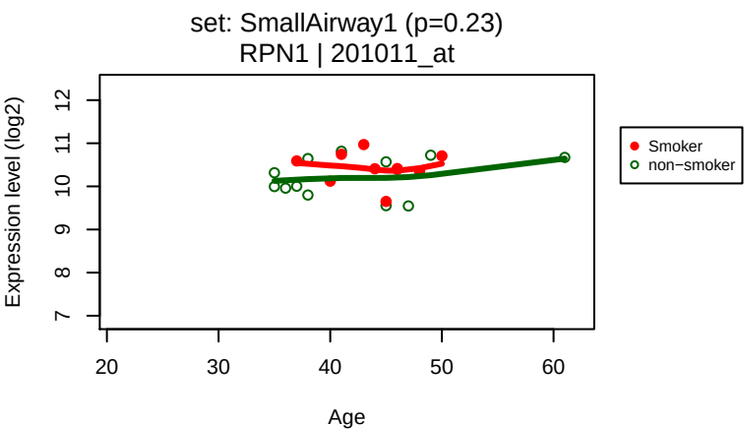
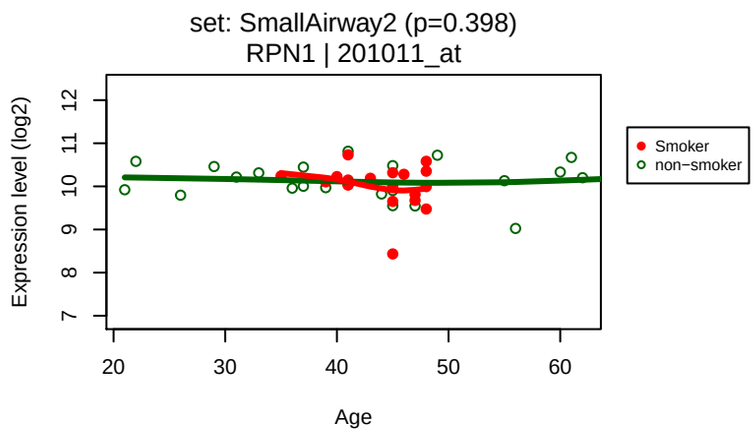
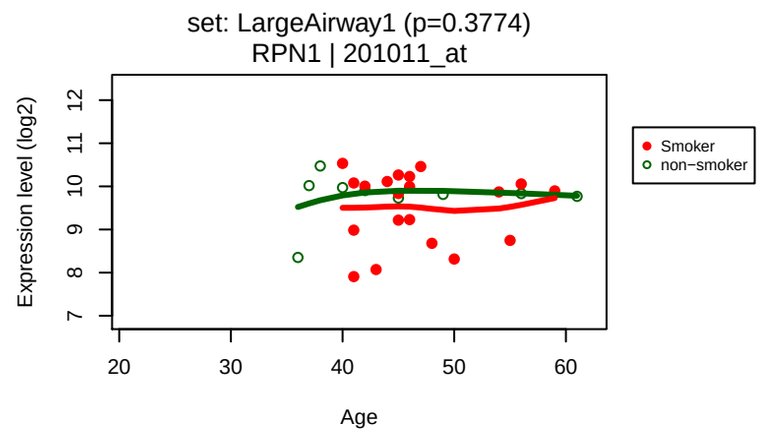

Gene expression of smokers and nonsmokers
gene: RPN1 | 201011_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.23
p=0.398
p=0.4761
p=0.3774
p=0.1679
p=0.3356
p=0.2657
p=0.1058
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

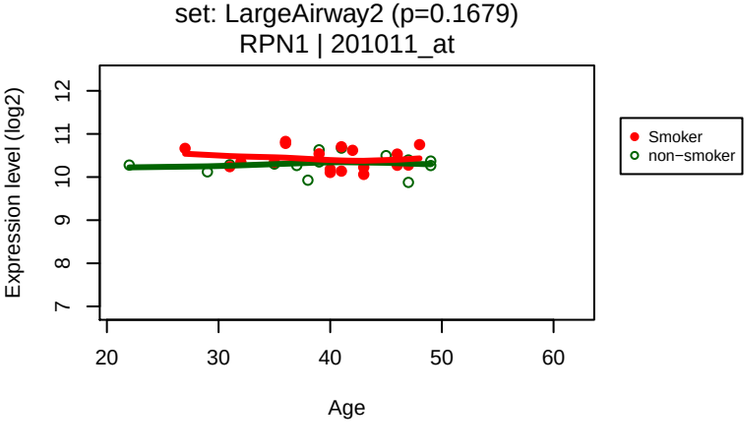

Gene expression ratio (smokers/nonsmokers)
gene: RPN1 | 201011_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.23
p=0.398
p=0.4761
p=0.3774
p=0.1679
p=0.3356
p=0.2657
p=0.1058
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

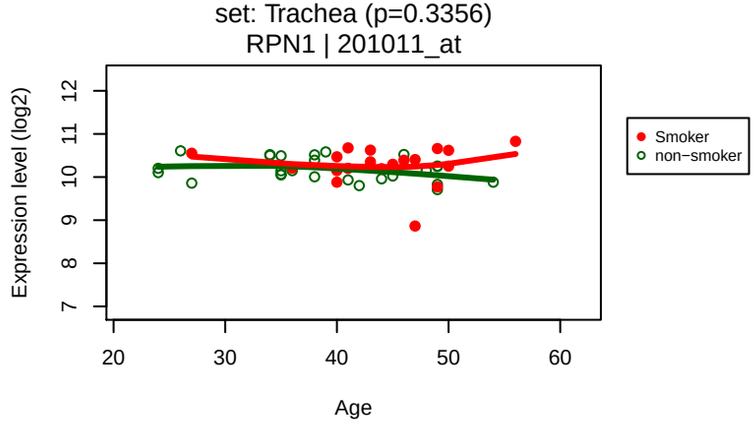

Gene expression ratio in Males
gene: RPN1 | 201011_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.569
p=0.3482
p=0.8623
p=0.044
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPN1 | 201011_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.55
p=0.2739
p=0.0113
p=0.9532
SmallAirway2
LargeAirway2
Trachea
Macrophage2

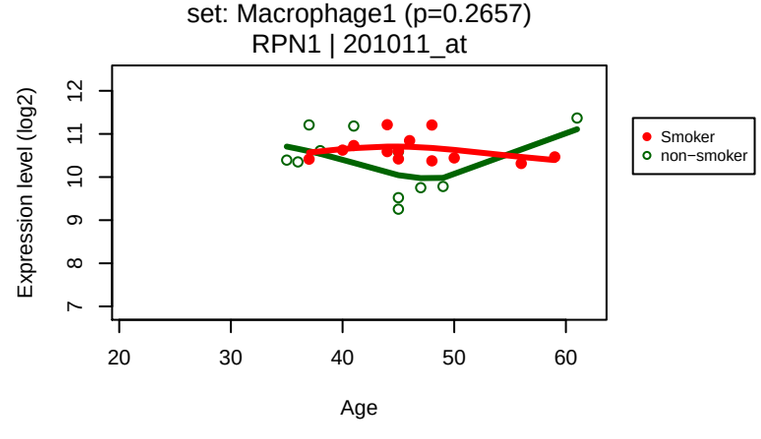

set: SmallAirway1 (p=0.23)
RPN1 | 201011_at
Expression level (log2)
Age
Smoker
non−smoker

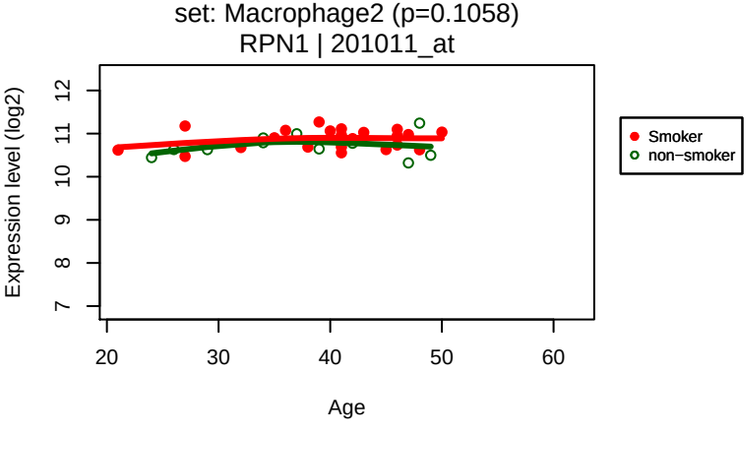

set: SmallAirway2 (p=0.398)
RPN1 | 201011_at
Expression level (log2)
Age
Smoker
non−smoker

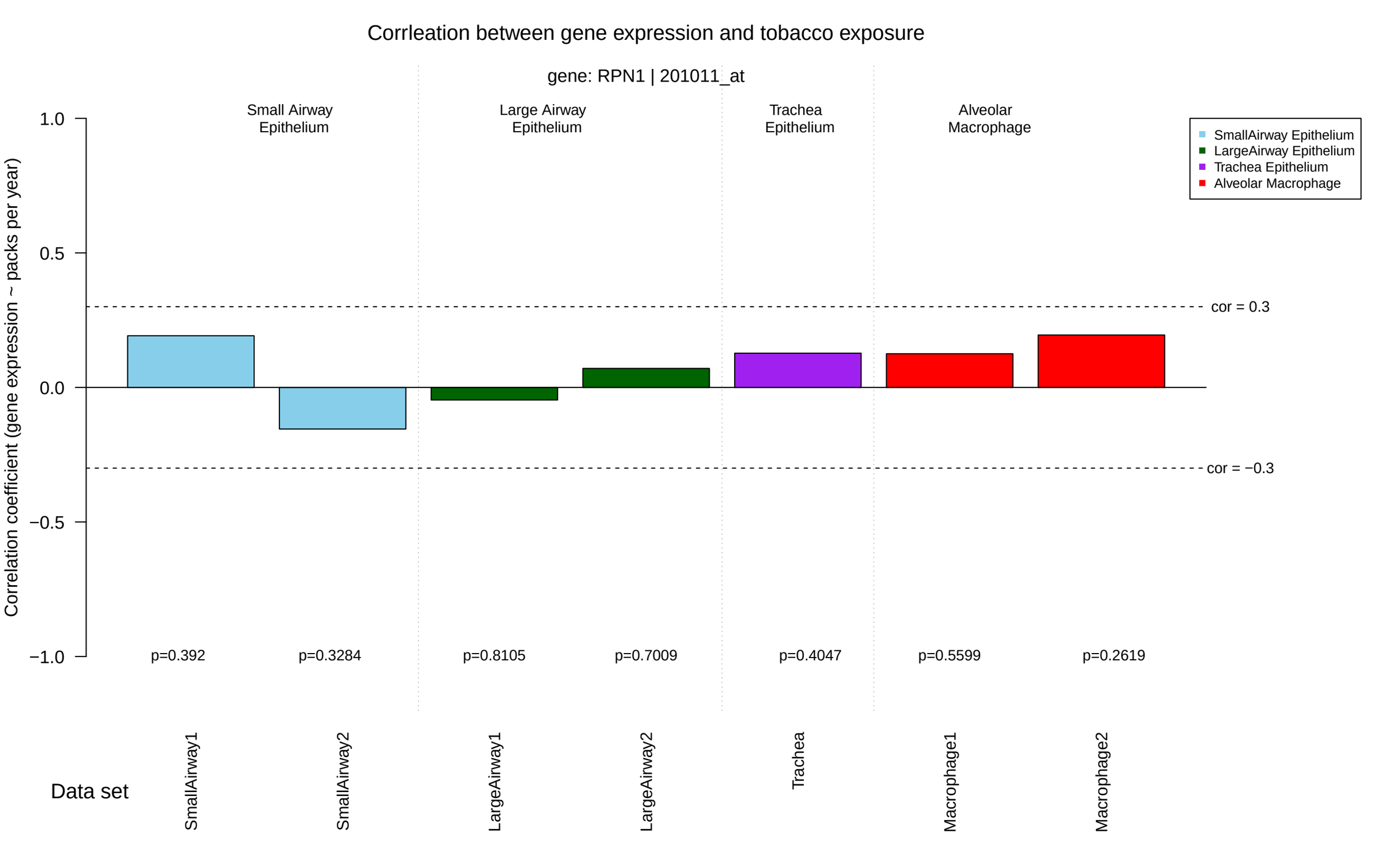

set: LargeAirway1 (p=0.3774)
RPN1 | 201011_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1679)
RPN1 | 201011_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3356)
RPN1 | 201011_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2657)
RPN1 | 201011_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1058)
RPN1 | 201011_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPN1 | 201011_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.392
p=0.3284
p=0.8105
p=0.7009
p=0.4047
p=0.5599
p=0.2619
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2